

1. Launch your IDE(MongoDB Atlas)
2. signup using Organization credentials
3. Create free tier cluster
4. If it is already created, RESUME it.
5. Click on Browse collections
6. Atlas Dashboard → Database → Browse Collections → Add My Own Data.
7. You're now inside MongoDB Atlas Data Explorer, and you have the sample_mflix database open with the comments collection selected. 8. To run CRUD operations,
 - a) Create (Insert)

DID the basic things through cmd prompt

Click "INSERT DOCUMENT" (top-right).

A JSON editor will open. Enter the
details in JSON

```
{  
  "name": "Alice",  
  "email": "alice@example.com",  
  "movie_id": "12345",  
  "text": "This is my first review!",  
  "date": "2025-08-23T10:00:00Z"  
}
```

Click Insert to save.

```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

Atlas atlas-7wh58m-shard-0 [primary] test> show dbs
sample_mflix 155.49 MiB
schoolDB      80.00 KiB
admin         0 B
local         0 B
Atlas atlas-7wh58m-shard-0 [primary] test> use schoolDB
switched to db schoolDB
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.insertOne({
  name: "Alice",
  age: 21,
  major: "Computer Science"
})
{
  acknowledged: true,
  insertedId: ObjectId('6a8930732d71267c8b95ac14')
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>
```

b) Read (Find)

In the Filter box, type queries in JSON format.

Example: Find comments by Alice:

{ "name": "Alice" } click Apply.

```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
acknowledged: true,
insertedId: ObjectId('6a8930732d71267c8b95ac14')
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.find({ major: "Computer Science" })
[
  {
    _id: ObjectId('6a867af0d44ce90bd13e0e62'),
    name: 'Alice',
    age: 22,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a867d1fd44ce90bd13e0e65'),
    name: 'David',
    age: 21,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a8930732d71267c8b95ac14'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science'
  }
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>
```

c)Update

Find the document first using a filter (e.g., { "name": "Alice" }).

Hover over the document → click pencil icon (Edit).

Modify fields, e.g. change "text" to "Updated review!".

Save it.

db.students.updateOne(

 { name: "Alice" },

 { \$set: { age: 22 } }

)

```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
},
{
  _id: ObjectId('6a8930732d71267c8b95ac14'),
  name: 'Alice',
  age: 21,
  major: 'Computer Science'
}
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.updateOne(
|   { name: "Alice" },
|   { $set: { age: 22 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>
```

D)Delete

Hover over the document you want to remove.

Click the trash bin icon (Delete).

```
db.students.deleteOne({ name: "Alice" })
```

```

mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
    major: 'Computer Science'
  }
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.updateOne(
|   { name: "Alice" },
|   { $set: { age: 22 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.deleteOne({ name: "Alice" })
{ acknowledged: true, deletedCount: 1 }
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>

```

2. Run Queries:

-> use schoolDB

Insert one document,

->db.students.insertOne({ name: "Alice", age: 21, major: "Computer Science" })

Find document,

->db.students.find({ major: "Computer Science" })

Update document,

->db.students.updateOne({ name: "Alice" }, { \$set: { age: 22 } })

Delete document,

->db.students.deleteOne({ name: "Alice" })

Indexing in MongoDB:

Indexes make searches much faster (just like an index in a book).

Insert many documents in the same database.

```

db.students.insertMany([
  { name: "Alice", age: 21, major: "Computer Science" },
  { name: "Bob", age: 22, major: "Mathematics" },

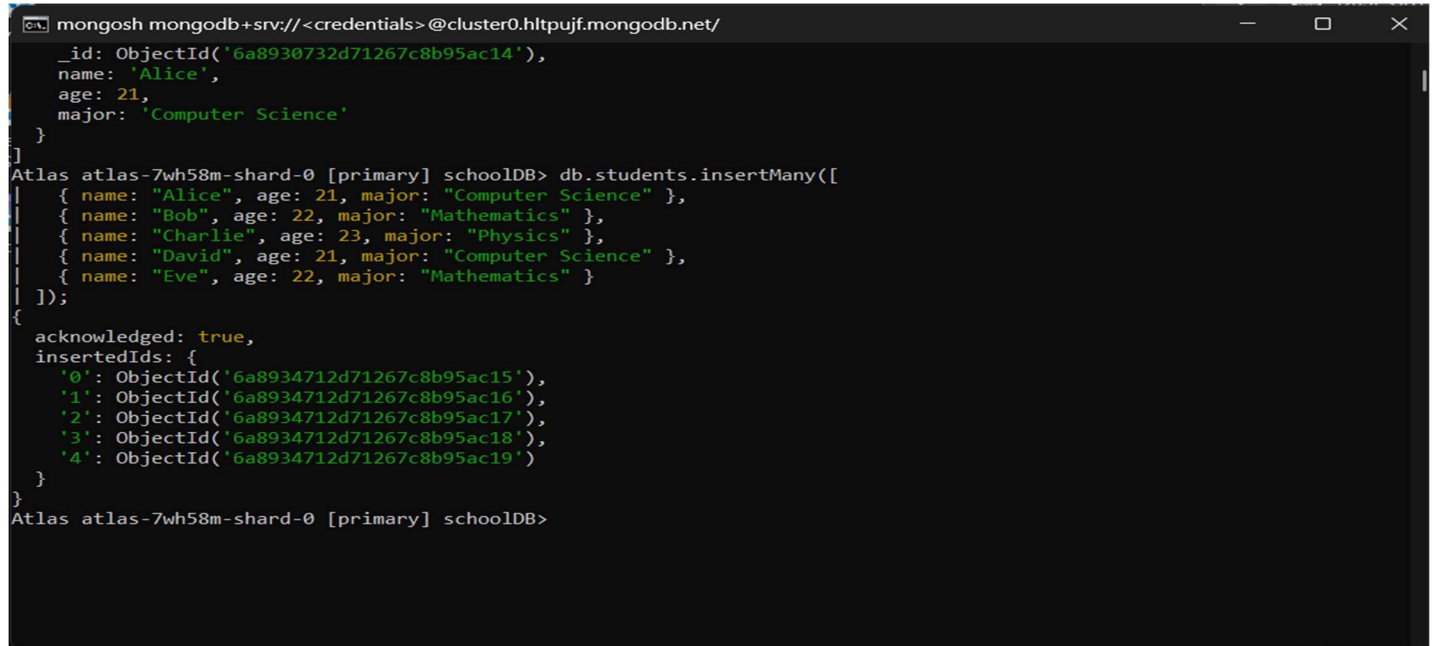
```

```

{ name: "Charlie", age: 23, major: "Physics" },
{ name: "David", age: 21, major: "Computer Science" },
{ name: "Eve", age: 22, major: "Mathematics" }
]);

```

2. Create an Index



```

mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
{
  _id: ObjectId('6a8930732d71267c8b95ac14'),
  name: 'Alice',
  age: 21,
  major: 'Computer Science'
}
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.insertMany([
  { name: "Alice", age: 21, major: "Computer Science" },
  { name: "Bob", age: 22, major: "Mathematics" },
  { name: "Charlie", age: 23, major: "Physics" },
  { name: "David", age: 21, major: "Computer Science" },
  { name: "Eve", age: 22, major: "Mathematics" }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8934712d71267c8b95ac15'),
    '1': ObjectId('6a8934712d71267c8b95ac16'),
    '2': ObjectId('6a8934712d71267c8b95ac17'),
    '3': ObjectId('6a8934712d71267c8b95ac18'),
    '4': ObjectId('6a8934712d71267c8b95ac19')
  }
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>

```

->db.students.createIndex({ major: 1 });

(1 = ascending, -1 = descending)

```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
{ name: "David", age: 21, major: "Computer Science" },
{ name: "Eve", age: 22, major: "Mathematics" }
});

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8934712d71267c8b95ac15'),
    '1': ObjectId('6a8934712d71267c8b95ac16'),
    '2': ObjectId('6a8934712d71267c8b95ac17'),
    '3': ObjectId('6a8934712d71267c8b95ac18'),
    '4': ObjectId('6a8934712d71267c8b95ac19')
  }
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.createIndex({ major: 1 });
major_1
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>
```

3. View Indexes

-> db.students.getIndexes();

```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
});
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8934712d71267c8b95ac15'),
    '1': ObjectId('6a8934712d71267c8b95ac16'),
    '2': ObjectId('6a8934712d71267c8b95ac17'),
    '3': ObjectId('6a8934712d71267c8b95ac18'),
    '4': ObjectId('6a8934712d71267c8b95ac19')
  }
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.createIndex({ major: 1 });
major_1
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { major: 1 }, name: 'major_1' }
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>
```

4. Compound Index

-> db.students.createIndex({ major: 1, age: -1 });

```
Select mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8934712d71267c8b95ac15'),
    '1': ObjectId('6a8934712d71267c8b95ac16'),
    '2': ObjectId('6a8934712d71267c8b95ac17'),
    '3': ObjectId('6a8934712d71267c8b95ac18'),
    '4': ObjectId('6a8934712d71267c8b95ac19')
  }
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.createIndex({ major: 1 });
major_1
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { major: 1 }, name: 'major_1' }
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.createIndex({ major: 1, age: -1 });
major_1_age_-1
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>
```

5. Drop Index

-> db.students.dropIndex("major_1"); Aggregation Framework

```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
{
  '2': ObjectId('6a8934712d71267c8b95ac17'),
  '3': ObjectId('6a8934712d71267c8b95ac18'),
  '4': ObjectId('6a8934712d71267c8b95ac19')
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.createIndex({ major: 1 });
major_1
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { major: 1 }, name: 'major_1' }
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.createIndex({ major: 1, age: -1 });
major_1_age_-1
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.dropIndex("major_1");
{
  nIndexesWas: 3,
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1787376969, i: 6 }),
    signature: {
      hash: Binary.createFromBase64('0Rj3J81+yUJgo9yjuYUdEY09fMk=', 0),
      keyId: Long('7636780945567121423')
    }
  },
  operationTime: Timestamp({ t: 1787376969, i: 6 })
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>
```

in MongoDB:

Count Students by Major

-> db.students.aggregate([
 { \$group: { _id: "\$major", total: { \$sum: 1 } } }
]);


```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.dropIndex("major_1");
{
  nIndexesWas: 3,
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1787376969, i: 6 }),
    signature: {
      hash: Binary.createFromBase64('0Rj3J81+yUJgo9yjuYUdEY09fMk=', 0),
      keyId: Long('7636780945567121423')
    }
  },
  operationTime: Timestamp({ t: 1787376969, i: 6 })
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.aggregate([
  { $group: { _id: "$major", total: { $sum: 1 } } }
]);
[
  { _id: 'Mathematics', total: 4 },
  { _id: 'Physics', total: 2 },
  { _id: 'Computer Science', total: 4 }
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>
```

Groups by major and counts students.

2. Average Age by Major

```
-> db.students.aggregate([
  { $group: { _id: "$major", avgAge: { $avg: "$age" } } }]);
```

```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.aggregate([
  { $group: { _id: "$major", avgAge: { $avg: "$age" } } }
]);
[
  { _id: 'Mathematics', avgAge: 22 },
  { _id: 'Physics', avgAge: 23 },
  { _id: 'Computer Science', avgAge: 21 }
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.aggregate([
  { $group: { _id: "$major", avgAge: { $avg: "$age" } } }
]);
[
  { _id: 'Physics', avgAge: 23 },
  { _id: 'Mathematics', avgAge: 22 },
  { _id: 'Computer Science', avgAge: 21 }
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>
```

3. sort by Age

```
-> db.students.aggregate([
  { $sort: { age: -1 } }
]);
```

```

mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
{ _id: 'Mathematics', avgAge: 22 },
{ _id: 'Computer Science', avgAge: 21 }
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.aggregate([
  { $sort: { age: -1 } }
]);
{
  _id: ObjectId('6a867d1fd44ce90bd13e0e64'),
  name: 'Charlie',
  age: 23,
  major: 'Physics'
},
{
  _id: ObjectId('6a8934712d71267c8b95ac17'),
  name: 'Charlie',
  age: 23,
  major: 'Physics'
},
{
  _id: ObjectId('6a867d1fd44ce90bd13e0e63'),
  name: 'Bob',
  age: 22,
  major: 'Mathematics'
},
{
  _id: ObjectId('6a867d1fd44ce90bd13e0e66'),
  name: 'Eve',
  age: 22,
  major: 'Mathematics'
}

```

4. Lookup (Join with another collection) create another collection

```

-> db.courses.insertMany([
  { course: "DBMS", major: "Computer Science" },
  { course: "Algebra", major: "Mathematics" },
  { course: "Quantum Physics", major: "Physics" }
]);

```

```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
{
  _id: ObjectId('6a8934712d71267c8b95ac18'),
  name: 'David',
  age: 21,
  major: 'Computer Science'
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.courses.insertMany([
  { course: "DBMS", major: "Computer Science" },
  { course: "Algebra", major: "Mathematics" },
  { course: "Quantum Physics", major: "Physics" }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8936a02d71267c8b95ac1a'),
    '1': ObjectId('6a8936a02d71267c8b95ac1b'),
    '2': ObjectId('6a8936a02d71267c8b95ac1c')
  }
}
Atlas atlas-7wh58m-shard-0 [primary] schoolDB>
```

Join with students:

```
-> db.students.aggregate([ {
  $lookup: { from: "courses",
  localField: "major", foreignField:
  "major", as: "enrolledCourses"
  }}
]);
```

```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.aggregate([
{
  $lookup: {
    from: "courses",
    localField: "major",
    foreignField: "major",
    as: "courseInfo"
  }
}
]);
[
  {
    _id: ObjectId('6a867d1fd44ce90bd13e0e63'),
    name: 'Bob',
    age: 22,
    major: 'Mathematics',
    courseInfo: [
      {
        _id: ObjectId('6a8936a02d71267c8b95ac1b'),
        course: 'Algebra',
        major: 'Mathematics'
      }
    ]
  },
  {
    _id: ObjectId('6a867d1fd44ce90bd13e0e64'),
    name: 'Charlie',
    age: 23,
    major: 'Physics',
    courseInfo: [

```

